

Ecological Emotions in a Climate-Vulnerable Nation: Eco-Anxiety, Eco-Guilt, and Eco-Grief Among Gen Z Filipinos

Marc Eric S. Reyes^{1,2} , Ma. Criselda T. Pacquing^{1,2}, Karen Katrina V. Trinidad^{1,2}, Jia Isabel C. Balmaceda¹, Aaron Jac J. Clemente¹, Andrea Clarisse M. De Jesus¹, Ruth Johanna A. Gool¹, Alekshia Quinne A. Isidro¹, and Charles A. Ogunbode³

Abstract

Ongoing global environmental change has given rise to a growing body of research on emotional and affective responses to ecological degradation. This study investigated eco-anxiety, eco-guilt, and eco-grief among Generation Z Filipinos, thereby contributing to the Global South perspective in a literature dominated by Western samples. Using a mixed-methods sequential explanatory (QUAN → qual) design, Study 1 employed a cross-sectional survey of 616 Gen Z Filipinos (aged 18–29) using the Eco-Anxiety Questionnaire, Eco-Guilt Questionnaire, and Ecological Grief Questionnaire. Study 2 involved qualitative interviews with nine participants. Quantitative findings revealed that participants exhibited high levels of habitual ecological worry, eco-guilt, and eco-grief, but relatively low functional impairment from eco-anxiety. Significant gender differences emerged, with females reporting higher levels of habitual ecological worry, eco-guilt, and eco-grief than males, although both genders experienced similar negative consequences. Place of residence also mattered: rural residents reported significantly greater negative consequences from eco-anxiety than urban residents, despite comparable levels of ecological worry, guilt, and grief. Qualitative analysis identified five themes—*Gendered Emotional Expression, Responsibility Rooted in Nurturing Roles, Emotional Resonance and Collective Grief, Proximity to Environmental Harm, and Disruption of Tradition and Place Identity*—that illuminate how gender socialization and environmental exposure shape ecological emotions. Our preliminary empirical insights and findings highlight the importance of recognizing ecological emotions as legitimate psychological phenomena warranting further investigations and attention in mental health practice and environmental education, particularly among climate-vulnerable youth populations.

Keywords

ecological crisis, eco-anxiety, eco-guilt, eco-grief, Gen Z, Filipinos

Received: 15 December 2025; Revised: 18 May 2026; Accepted: 27 May 2026

¹Department of Psychology, College of Science, University of Santo Tomas, Manila, Philippines

²Research Center for Social Sciences and Education, University of Santo Tomas, Manila, Philippines

³School of Psychology, University of Nottingham, Nottingham, UK

Corresponding author:

Marc Eric S. Reyes, Department of Psychology, College of Science, University of Santo Tomas, España Boulevard, Sampaloc, Manila, 1008 Metro Manila, National Capital Region, Philippines.

Email: msreyes@ust.edu.ph



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

Introduction

The escalating ecological crisis, marked by interconnected issues such as climate change, biodiversity loss, and environmental degradation, poses a severe global threat.¹ This has been described by the United Nations Secretary-General, António Guterres, as a “code red for humanity”.² Importantly, the global ecological crisis is generating profound psychological impacts and giving rise to new forms of environmental distress including feelings of grief, loss, and frustration.³ New terminology is being rapidly developed to capture these specific psychological responses to environmental degradation. Associated with this effort is a burgeoning literature on “ecological emotions”, which refer to the emotional impact of environmental degradation and climate change and which are often described using terms associated with psychopathology.⁴ Some of the most prominent constructs in this emerging area of research are: “eco-anxiety,” “ecological guilt,” and “environmental grief.”

Within the Affective-Existential Framework of Climate Distress, eco-anxiety, ecological guilt, and environmental grief are understood as central emotional processes that both constrain and motivate human adaptation.⁵ Rather than viewing climate-related distress as a purely clinical fear response, this framework conceptualizes it as an existential threat to core dimensions of human life—mortality, meaning, isolation, and freedom—thereby giving rise to a range of ecological emotions, including fear, anger, grief, and hope.

While cognitive reactions measure understanding and behavioral reactions measure outward action, they often fail to capture the “internalized crisis” that determines long-term psychological resilience.³ These three variables are uniquely sensitive to social-ecological positioning: environmental grief captures the loss of place-based identity,⁶ ecological guilt captures the conflict of modern agency,⁷ and eco-anxiety captures the anticipatory dread of systemic collapse.⁸ Moreover, these constructs are fundamentally psychological reactions to the ecological crisis, marked by corresponding emotional states that are subsequently interconnected with cognitive patterns and behavioral manifestations.^{9–11} As such, highlighting eco-emotions allows for specificity in relation to the ecological context, marking its distinction from broader, encompassing psychological reactions.

Consequently, focusing on this triad provides a more comprehensive view of the ‘lived experience’ of the ecological crisis than a focus on purely cognitive or behavioral metrics. Thus, this present study seeks to comprehensively investigate and provide an in-depth understanding of the psychological responses of individuals in a climate-vulnerable nation to major ecological challenges by integrating quantitative and qualitative data using a mixed-methods design structured within a sequential explanatory framework.

Defining Anxiety, Guilt, and Grief in the Context of Ecological Crisis

Eco-anxiety is often used interchangeably with “climate anxiety”. However, eco-anxiety encompasses a range of adverse emotions, such as worry, fear, anger, and despair, in response to the ecological crisis,^{5,7,12} while climate anxiety specially captures anxiety regarding the climate crisis.^{7,12} It is also crucial to differentiate eco-anxiety, which can involve clinically significant suffering, from mild normative worry.¹³

Alongside anxiety, researchers also explored feelings of guilt arising from the ecological crisis. “Ecological guilt” refers to the feeling of regret, a sense of accountability, and uneasiness when one recognizes their role in environmental damage, whether intentional or unintentional.¹⁴ Ágoston and colleagues⁹ found that eco-guilt was linked to collective guilt, thus, making people depend on group efforts to solve ecological issues. Furthermore, they mentioned that eco-guilt encompasses feelings of regret regarding personal actions and choices, such as the over-production of waste that causes harm to the natural world. Specific literature has identified some effects of eco-guilt on individuals.^{9,15} Ágoston and colleagues⁹ found that eco-guilt can stimulate personal responsibility and self-criticism, resulting in greater awareness of one’s influence on the environment. Thus, this cultivates a desire within the individual to modify their values and actions as guided by environmental principles.

Following eco-anxiety and eco-guilt, another related prominent emotion in the literature is ecological grief. “Ecological grief” refers to the perceived sense of loss in the environment or an anticipated loss in the future as a result of a deteriorating environment.¹⁴ Benham and Hoerst¹⁶ further elaborated that this ecological emotion is characterized by a sense of yearning, sorrow, and sadness in response to their environment’s past, present, or future ruination. These feelings could be manifested in a variety of contexts; thus, three types of eco-grief have been identified: grief related to physical and ecological loss, grief related to the loss of the environmental knowledge system and loss of identity, and grief associated with anticipated ecological loss.¹⁷ Eco-grief has been linked to acute events and chronic environmental losses.¹⁶ In addition, Rai mentioned that eco-grief was prevalent among individuals who have a close relationship with nature and the environment around them.¹⁷ Other studies^{14,16} support this observation, finding that many people possess place attachment and a sense of belongingness with their environment.

Locating the Global South in Ecological Emotions Research

Current research on ecological emotions remains limited by its focus on WEIRD (Western, Educated, Industrialized, Rich, and Democratic) populations, neglecting a global context. This is a critical oversight, as the Global Environmental Crises (GEC) disproportionately impact impoverished and disadvantaged communities. A 2023 study by Voški and colleagues in Turkey identified a pattern of distinct ecological emotions, driven by factors like environmental justice, developing country tension, self-efficacy, and responsibility attribution.¹⁸ Contrary to the pervasive eco-anxiety found in WEIRD nations, their research found eco-anger to be the most prominent emotion among Turkish participants.

The disproportionate scholarly attention to global environmental challenges of the Global North represents a critical research gap, as the psychological impacts of ecological crises differ markedly for the Global South. For instance, Tsevreli and colleagues found that eco-anxiety varies in both degree and intensity between these regions.¹² Furthermore, Stafford and colleagues note that psychological struggles in the Global South are compounded by factors like political instability, conflict, and limited access to mental health services.¹⁹

These compounded pressures create severe psychological distress and heightened vulnerability. A recent international poll⁸ reflects this, showing that a large majority of young people in Global South nations, including the Philippines (92%), Brazil (86%), India (80%), and Nigeria (70%), report intense fear of climate change. Notably, this and other studies consistently identify anger, not anxiety, as the most prevalent ecological emotion in these contexts. Hickman and colleagues⁸ also reported anger as a primary response, a finding corroborated by Ogunbode et al.²⁰ and supported by the qualitative work of Kleres and Wettergren on environmental activists.²¹

Ecological Emotions in the Philippines

The Philippines is uniquely vulnerable to the impacts of climate change. As an archipelago, its geographical location and physical climate make it susceptible to a multitude of disasters caused by natural hazards, including typhoons, floods, earthquakes, and landslides.²² It is consistently ranked among the top three nations most affected by such events, facing an average of eight calamities and as many as 20 typhoons annually.²³⁻²⁵ This high level of exposure has positioned the country as being at significant risk from climate change.²⁴ Consequently, an international poll by Hickman and colleagues found that the Philippines had the highest proportion of participants (92%) among Global South nations who feared the effects of climate change on their future.⁸

This constant threat has a profound psychological impact, particularly on the younger generation. While Filipinos in disaster-prone areas show greater climate change awareness and engagement in adaptation,²⁴ they also develop more intense negative emotions. Studies show that Filipino youth experience heightened worry that affects their daily functioning.¹² The Philippines has the highest rate of young people experiencing intense climate anxiety, as well as emotions like anger, hopelessness, and frustration.²⁴ Tsevreli and colleagues strengthen this finding, reporting that 83% of the younger generation in the Philippines experience eco-anxiety, with 62.4% experiencing a particularly dark, intense, and impactful form of this affective response.¹²

The focus on youth is critical globally as Generation Z, often called the “climate generation”, has spent their entire lives affected by environmental adversities.¹² Living through rapid climate change and ecosystem loss, this generation is considered one of the most predisposed to its adverse psychological effects.²⁶ They hold a dark picture of the environmental future²⁷ and express significant worry about the long-term effects of climate change they will inevitably experience.^{24,28} This distress manifests as ecological emotions; from 2010 to 2019, feelings of worry, anger, and guilt regarding climate change increased among younger generations.²⁹ Current literature confirms a higher prevalence of eco-anxiety in young individuals, with Gen Z displaying strong feelings of guilt due to an inability to protect the environment.^{12,14} Furthermore, when their concerns are disregarded, it can aggravate feelings of isolation and distress, particularly for those who feel powerless to enact change.³⁰

The Current Study

Despite growing research on the psychological effects of the ecological crisis, significant gaps remain, especially in the Philippine context. There is a need for an extensive investigation focusing on the experiences of anxiety, guilt, and grief among Gen Z Filipinos. Most existing studies focus on a single emotion^{7,12,15,28,31-33} and lack profound knowledge of how the interplay of these emotions affects the youth. Pihkala also identified a scarcity of studies on the different manifestations of anxiety and their relationship with other psychological impacts.⁷ Furthermore, in spite of the repeated occurrence of disasters caused by natural hazards in the Philippines, overall public awareness of climate change remains low.²⁸ This study aims to

address these gaps by supplying more specific information and exploring how emotional responses to the ecological crisis relate to demographic variables, such as regional residence, and mental health, as suggested by Reyes and colleagues.²⁸

Our objective is supported by the theory of Ontological Security, introduced by psychiatrist R.D. Laing in 1960 and reinvented by sociologist Anthony Giddens in 1991. This theory emphasizes that feelings of security or insecurity deeply influence an individual's well-being and behavior, supporting the claim that environmental problems can result in ontological insecurity that develops into anxiety, depression, or defensive emotions.⁷ This study focused on Gen Z Filipinos, born between 1995 and 2010³⁴ and aged 18 to 29, as this generation exhibits significant stress about future concerns like climate change²⁸ and is highly vulnerable to its consequences despite negligible contribution to causing the crisis.²⁶

Moreover, testing gender differences was necessary because previous literature (e.g., Clayton,³²) suggests women often carry a disproportionate emotional burden regarding climate change. Thus, the present study aimed to see if this Global North trend held true in a Global South context like the Philippines. Likewise, gender (i.e., male and female) and location (i.e., urban and rural) are key variables for understanding eco-anxiety, eco-guilt, and eco-grief.⁶ To understand why demographic and geographic factors are not just 'check-boxes' but core drivers of environmental emotion, we have to look at the Social-Ecological Vulnerability Framework.³⁵ This framework suggests that eco-anxiety, guilt, and grief are not distributed equally because our social positions dictate how much we have to lose and how much responsibility we feel forced to carry.³² We cannot treat environmental distress as a 'one-size-fits-all' clinical issue. It is a rational response to one's specific social and geographical reality.³⁶ This framework illustrates how individual vulnerability is not static but is nested within micro (family/local environment), meso (community), and macro (culture/government) systems that dictate emotional responses to environmental change.^{37,38}

To delve into their psychological responses, we employed a mixed-methods sequential explanatory approach (QUAN → qual), responding to the call by Tseveni and colleagues¹² to explore eco-anxiety through both quantitative and qualitative research in climate-affected countries. Using a sequential mixed-methods design, the study first conducted a descriptive, cross-sectional quantitative assessment of ecological emotions, followed by a descriptive qualitative inquiry aimed at contextualizing and elaborating on Gen Z Filipinos' experiences of eco-anxiety, eco-guilt, and eco-grief.

Method

A mixed-methods sequential explanatory design (QUAN → qual) was utilized to investigate eco-anxiety, eco-guilt, and eco-grief among Generation Z Filipinos. Quantitative data were collected and analyzed first in Study 1, followed by a qualitative inquiry intended to contextualize and deepen interpretation of the quantitative results in Study 2.

The present investigation secured ethics approval from Davao de Oro State College – Research Ethics Committee (Protocol Code: 824-08-2024). We sought participants who are Filipinos belonging to Gen Z, aged 18 to 29 years old. Participants who were experiencing psychological distress or had a diagnosed psychological condition at the time of data collection were excluded from the study. This criterion was applied to ensure that reported ecological emotions (i.e., anxiety, guilt, and grief) were attributable primarily to the ecological crisis rather than to pre-existing psychological vulnerabilities. Ethically, this decision also reflected the sensitive nature of the topic and its potential to exacerbate distress among individuals with psychological concerns.

Study 1

Design

A descriptive, cross-sectional research design was employed using a self-report questionnaire to collect data at a single point in time³⁹ (Johnson, 2001). The implementation of the descriptive, cross-sectional design examined the anxiety, guilt, and grief that Gen Z Filipinos have in relation to the ecological crisis. This approach aimed to provide further description and understanding of how the ongoing ecological crisis affects the ecological emotions of the specific population, thereby capturing their emotional profiles.^{40,41}

Participants

We utilized the G*power 3.1.9.7 analysis with reference to a medium effect size of Cohen's $d = 0.5$ and arrived at a target sample of at least 210 participants. We aimed to describe the statistics derived from the collected participants data and analyze how ecological emotions persist and become more pronounced in specific subgroups of participants, particularly gender and the area of residency being a point of consideration. Gender identity and the area of residency were reported to have correlations with experiencing ecological emotions according to the literature review.^{9,42-48} Hence, these characteristics are

deemed relevant and will be considered during our data collection. Out of the 747 survey participants, 131 cases were excluded based on the screening questions (118 participants answered Yes to: Are you experiencing any distress at the moment that may significantly affect you from completing an online survey?; 34 participants answered yes to: Are you diagnosed with any mental health disorder that may significantly affect you from completing an online survey?). Table 1 presents the demographic profile of the final sample comprising 616 Generation Z Filipino participants aged 18–29 years ($M = 21.73$, $SD = 2.60$). All participants met the inclusion criteria: (1) Filipino individuals residing in the Philippines at the time of data collection; (2) aged between 18 and 29 years; and (3) possessing sufficient proficiency in English to comprehend the survey materials.

Measures

Eco-Anxiety Questionnaire (EAQ-22)

Eco-Anxiety Questionnaire (EAQ-22) is a newly developed questionnaire by Ágoston and colleagues.⁵ It consists of 22 items, answered by a 4-point Likert scale (1 = Strongly disagree, 2 = Somewhat disagree, 3 = Somewhat agree, and 4 = Strongly agree) and evaluates two factors: habitual ecological worry and negative consequences of eco-anxiety. The habitual ecological worry factor consists of items (items 1, 2, 4, 6, 7, 9, 11, 12, 14, 15, 17, 19, 20) that primarily address external concerns unrelated to the effects of anxiety on an individual, including displays of strong emotions (e.g., being scared, alarmed, terrified, or worried about the consequences of climate change and the ecological crisis) and uncertainty of the future and disturbing changes (e.g., changing seasons, disasters caused by natural hazards). Additionally, within this factor are items that indicate concern for others or anger towards the negative environmental behaviors of others (individuals, leaders, countries). A sample item is “*I am worried about the increasing number of natural disasters caused by climate change.*” The negative consequences of eco-anxiety factor (items 3, 5, 8, 10, 13, 16, 18, 21, 22) indicate concrete physical, emotional, or behavioral effects (e.g., poor sleep, crying, constraint alertness), eco-paralysis or the inability to effectively address ecological issues,⁴⁹ and functional and social impairment. A sample item is “*I am so anxious about climate change that I cry.*” The habitual ecological worry scale shows a high internal consistency with a Cronbach’s alpha of .91, while the negative consequences of eco-anxiety scale also show high internal consistency, Cronbach’s alpha of .86.⁵ In the present study, the EAQ-22 has the following Cronbach’s alphas: Habitual Ecological Worry ($\alpha = .90$) and Negative Consequences ($\alpha = .89$).

Eco-Guilt Questionnaire (EGuiQ-11)

A questionnaire developed by Ágoston and colleagues⁵ consists of 11 items comprising different forms of eco-guilt (e.g., blaming oneself, discontent with one’s action, guilt over the behavior of one’s friend or family, involvement in an environmentally harmful system, and existential guilt). It is a unidimensional questionnaire wherein a single factor is being measured. A high score on this questionnaire suggests a stronger presence of eco-guilt. Respondents answer through a 4-point Likert scale (1 = Strongly disagree, 2 = Somewhat disagree, 3 = Somewhat agree, and 4 = Strongly agree). Some sample items in this questionnaire include, “*I feel guilty for not paying enough attention to the issue of climate change*” and “*The more I know about the human causes of climate change, the more things I feel guilty about.*” The Cronbach’s alpha ($\alpha = .89$)

Table 1. Profile of the Participants in Study 1 (N = 616)

Characteristics	n (%)
Gender	
Male	196 (31.82%)
Female	420 (68.18%)
Place of Residence	
Urban	463 (75.16%)
Rural	153 (24.84%)
Have you experienced any significant environmental calamity/calamities in the last three months?	
Yes	457 (74.19%)
No	159 (25.81%)

of the scale for the total sample indicates an excellent internal consistency.⁵ The EGuiQ-11 in our study has a reliability coefficient of .88.

Ecological Grief Questionnaire (EGriQ-6)

A six-item questionnaire developed by Ágoston and colleagues⁵ that measures general ecological grief. It examines the feelings of loss brought on by wildlife changes, grief over local and remote habitat destructions, and uncertainty about the consequences of these changes.⁵ The questionnaire is answered by a 4-point Likert scale (1 = Strongly disagree, 2 = Somewhat disagree, 3 = Somewhat agree, and 4 = Strongly agree). It is a unidimensional questionnaire wherein a high score indicates more pronounced eco-grief. A sample item is “*I feel some sense of loss because of climate change impacts that are becoming apparent in my local area.*” The scale was observed to have an excellent internal consistency with a Cronbach’s alpha of .77.⁵ The EGriQ-6 in our study has a reliability coefficient of .71.

The high internal consistency observed across all three scales—Eco-Anxiety ($\alpha = .90$ for habitual worry; $\alpha = .89$ for negative consequences), Eco-Guilt ($\alpha = .88$), and Ecological Grief ($\alpha = .71$) provides critical psychometric support for their use in this study. These robust reliability coefficients suggest that the instruments maintained their conceptual integrity and internal structure within a Filipino sample, offering empirical evidence of their suitability for assessing ecological emotions in this specific population even in the absence of formal cultural adaptation.

Procedure

Study 1 used voluntary response sampling by disseminating the questionnaires through an accessible link, which was posted and shared across various social media platforms such as Facebook, Messenger, X, Instagram, and TikTok. Data collection was done using an online research questionnaire via Google Forms survey. This survey contained information on the nature and rationale of the study, demographic questions, informed consent, and the test battery (comprises EAQ-22, EGuiQ-11, & EGriQ-6). The three scales in the test battery were randomized in order using the website [allocate.monster](https://www.allocate.monster/) to minimize and control the possible effects of systematic order. Study 1 used the original English scales without translation or cultural modification. This decision was based on the high level of English proficiency in the Philippine academic and professional sectors. To further mitigate potential language barriers, only individuals who self-identified as having a fair understanding of English were eligible for inclusion in the study. While the original English versions of the instruments were utilized due to the high English proficiency of the target academic demographic, the internal reliability of these measures was closely monitored to ensure their validity. The resulting Cronbach’s alpha scores, which are comparable to or exceed those reported in the original validation study by Ágoston and colleagues,⁵ serve as a form of sample-specific validation, indicating that the items were interpreted consistently by the participants. Upon completion of the online survey, participants received a detailed written debriefing alongside contact numbers for crisis hotlines (e.g., National Center for Mental Health hotline).

Data Analysis

The data gathered was subjected to descriptive and inferential statistical analysis. We used a web-based spreadsheet application Google Sheets (web version) and the statistical program Jamovi ver. 2.5.2 in performing the data analysis. Descriptive statistics (means and standard deviations) and independent-samples *t* tests were used in the present study. Welch’s *t* test was employed to account for unequal sample sizes, as it adjusts the degrees of freedom for unequal variances and provides strong control of Type I error.⁵⁰

Study 2

Design

A qualitative descriptive or QD design by Sandelowski⁵¹ was employed to obtain a deeper understanding of Gen Z Filipinos’ experiences regarding the ecological crisis, as this method is particularly suited for providing a comprehensive summary of an event or phenomenon in everyday language.⁵¹ A qualitative approach using thematic analysis within a sequential explanatory design was conducted using semi-structured interviews among a subset of participants identified in Study 1 to further explore their experiences and perspectives. This method of gathering data ensures an extensive and detailed comprehension of the viewpoints held by Gen Z Filipinos on the ecological crisis and its emotional impacts,⁵² which, while typically exploratory in nature, it enables an in-depth examination of participants’ lived experiences and provides richer explanation and contextualization of the quantitative findings.

Participants

Given the descriptive qualitative approach of Study 2, nine participants were selected, consistent with Creswell's recommendation that qualitative studies include a minimum of five and a maximum of 25 participants.⁵³ We opted to have at nine interviewees, considering that our target participants are a homogenous sample. This aligns with Morse's recommendation of at least six participants for qualitative studies.⁵⁴ Furthermore, in the study of Guest and colleagues, they reported basic elements for metathemes have already emerged by the sixth interview.⁵⁵ From the quantitative Study 1 participant pool ($N = 616$), 127 individuals indicated willingness to participate in interviews. As such, for our Study 2, nine participants were subsequently selected from this subgroup ($n = 127$) using purposive-random sampling—randomly selecting from within specific subgroups (e.g., rural-male, urban-female) to ensure the qualitative data could actually “explain” the differences found in Study 1 and to ensure diversity while maintaining methodological rigor. Although the target population was relatively homogeneous, we deliberately considered variation in key demographic characteristics—particularly gender (male & female) and place of residence (urban & rural)—because these variables emerged as relevant in the quantitative findings of Study 1. This approach allowed the qualitative investigation to capture a broader range of perspectives. Table 2 shows the characteristics of the nine participants in terms of residence and gender.

Interview Protocol

Robotfoto, personal data sheets that serve as a tool to characterize the demographic profile of the research participants⁵⁶ and it is utilized to confirm that the participants meet the inclusion criteria.⁵⁷ We particularly gathered the participant's age, gender, and their area of residence (whether rural or urban area).

Aide Memoire, a semi-structured interview was developed and served as a written aid of the interview process for both the researchers and the respondents.^{57,58} The interview guide included process-oriented questions that revolved around the lived experiences of the participants,⁵⁸ particularly, Gen Z Filipinos. The aide memoire was guided to seek substantial answers to the central question of the study (*How do Gen Z Filipinos experience and make sense of their eco-anxiety, eco-grief, and eco-guilt in the context of their gender and place of residence?*) that was based on the quantitative results of Study 1.

Procedure

The qualitative sample was drawn from participants in Study 1 who completed the initial questionnaire and indicated their willingness to participate in a subsequent online interview. Before conducting the interviews, the nine participants were provided with an informed consent form containing the necessary information regarding the present study. Each interview was held online via Zoom or Google Meet, depending on the participant's preference and convenience. Each interview lasted approximately one hour and 30 minutes to two hours. Minimizing harm, unnecessary information, and the use of vulgar language, as well as maximizing the safety and well-being of the participants, were observed as a high priority throughout the conduct of the study. With this, the privacy and comfort of the participants was ensured at all times. Consent in recording the interview was sought from each participant before the session. A token of appreciation (e.g., gift certificate) was provided to the interviewees in gratitude for their participation in the study. Lastly, all interview recordings were deleted after the completion of data transcription and thematic analysis.

Study 2's sample size composition ($N = 9$ in Table 2) reflects the purposive-random selection of participants from those who expressed willingness to participate in the qualitative investigation, with consideration given to variation in gender and

Table 2. Participant Demographics in Study 2 ($N = 9$)

Participant	Area of residence	Gender
P1	Rural	Male
P2	Urban	Female
P3	Urban	Male
P4	Rural	Female
P5	Urban	Male
P6	Urban	Male
P7	Rural	Male
P8	Urban	Female
P9	Rural	Female

residential contexts. Participants were drawn from the larger quantitative pool of 127 Gen Z Filipinos who completed the initial survey and indicated willingness to participate in interviews. The selection of nine participants aligns with qualitative research guidelines recommending a small, homogenous sample to enable in-depth exploration of lived experiences while ensuring data saturation. In qualitative research, the saturation point is reached when the essence of participants' lived experiences is thoroughly understood without the emergence of new information. Data saturation, according to Ahmed, is a pivotal concept in qualitative research wherein no new themes, insights, or patterns emerge, indicating that data collection can be halted.⁵⁹ This concept is integral not only in determining an adequate sample size but also contributes to the validity and credibility of qualitative research.⁵⁹

Data Analysis

Thematic analysis (TA) was used to understand experiences, thoughts, and behaviors across a data set.⁶⁰ Braun and Clark have identified six key steps in conducting TA.⁶¹ First, during the familiarization phase, we engaged and immersed ourselves in the data by thoroughly reading and rereading it to gain in-depth understanding. This is followed by the coding process, where we systematically assigned labels or codes to the significant aspects of the data, either through inductive reasoning or by utilizing a pre-existing coding framework. These initial codes were then organized into broader categories or themes by identifying the patterns and relationships present. We then reviewed and refined these themes to ensure that they accurately reflect the data. This was followed by defining and naming those themes, creating clear and distinct definitions for each. Here, we identified the essence of each theme and determined the specific aspects of data the theme captures. Finally, in the write-up phase, we provided a concise, coherent, logical, non-repetitive, and interesting report of the data, which ensured that the analysis was credible and valid.

Furthermore, we employed auditing and member checking as validation strategies to ensure the integrity of the qualitative analysis as well as to control possible confounding of results. Implementing an audit trail involves recording how a qualitative study was conducted and how the researchers arrived at their conclusions.⁶² Member checking on the other hand, involved presenting each interviewee with their own data transcripts and analysis to gather feedback, which provided credibility to the interpretation.⁶³

Integrating reflexivity was also essential for our second phase analysis, given the potential subjectivity and biases that may surface when analyzing the data gathered from our participants. Reflexivity, as defined by Gouldner in 1971, is the analytical focus on the researcher's role in qualitative research, which requires a high degree of self-awareness at every step of the research process.⁶⁴ It involves analyzing one's presumptions, beliefs, and judgments, and critically assessing how this influences the study.⁶⁵ Through reflexivity, we assessed how our experiences change as a result of the research process and evaluate how these changes affect the study itself.⁶⁴ Therefore, we recognized that the codes we input during TA represented our interpretations of the patterns seen in the dataset. We reflected on our identities as researchers and acknowledged how our values, life circumstances, social roles, and ideologies were shaped by the research we presented.⁶⁶ Through this, we adopted a critical thinking approach that challenges us to examine the 'whys' and 'hows' of our research, questioning the utility, ethics, and value of what, who, and how we study.^{65,67}

Results

Study 1: Quantitative

We observed a high level of eco-anxiety, eco-guilt, and eco-grief among our sample of Gen Z Filipinos by running one-sample t-tests to determine whether the participants' scores in ecological emotions were significantly different compared to the midrange scores considered as normal. As shown in Table 3, our results revealed that participants' eco-anxiety on the EAQ-22 Habitual Ecological Worry (HEW) subscale was significantly higher ($M = 44.3$, $SD = 5.91$) than the normative

Table 3. Comparison of Eco-Anxiety, Eco-Guilt, and Eco-Grief Levels Across the Sample (N = 616)

Ecological emotions	<i>M</i>	<i>SD</i>	Midrange score	Mean difference	<i>t</i>	<i>p</i>
Eco-anxiety (HEW)	44.3	5.91	32.5	11.8	49.7	<.001
Eco-anxiety (NC)	19.6	5.75	22.5	-2.89	-12.5	<.001
Eco-guilt	30.5	6.00	27.5	2.98	12.3	<.001
Eco-grief	18.3	2.97	15	3.34	27.9	<.001

Note. Significant *p* values appear in bold.

midrange HEW score of 32.5 (HEW: 13 to 52), with a mean difference of 11.8, $t(615) = 49.7, p < .001$. On the contrary, their eco-anxiety as measured by the EAQ-22 Negative Consequences [NC] subscale was lower ($M = 19.6, SD = 5.75$) compared to the midrange NC score of 22.5 (NC: 9 to 36), yielding a mean difference of -2.89, $t(615) = -12.5, p < .001$.

Table 3 also shows that our participants eco-guilt as measured by EGuiQ-11 was higher ($M = 30.5, SD = 6.00$) than the midrange score of 27.5 (EGuiQ-11: 11 to 44), resulting in a mean difference of 2.98, $t(615) = 12.3, p < .001$. Likewise, their eco-grief as measured by EGriQ-6 was higher ($M = 18.3, SD = 2.97$) compared to the midrange score of 15 (EGriQ-6: 6 to 24), with a mean difference of 3.34, $t(615) = 27.9, p < .001$.

Given the unequal gender distribution in the sample (420 females and 196 males), Welch's t-tests were conducted to examine differences in eco-anxiety, eco-guilt, and eco-grief across gender (male vs. female) and place of residence (urban vs. rural). Moreover, gender was identified as a key demographic variable in our investigation since it has been associated with ecological emotional responses in previous research (see 9, 43, 47). Furthermore, to account for multiple comparisons across the four t-tests used (two for the eco-anxiety subscales, one for eco-guilt, and one for eco-grief) and to reduce the risk of Type 1 error, a Bonferroni correction was utilized. The initial alpha value of $\alpha = .05$ was divided by the number of comparisons ($n = 4$), arriving at an adjusted significance level of $p < .0125$.

Table 4 shows that females had significantly higher eco-anxiety ($M = 44.9, SD = 5.71$) than males ($M = 43.1, SD = 6.16$) in the HEW subscale, $t(356) = -3.41, p = < .001$. However, no significant difference was found between the eco-anxiety of males ($M = 19.8, SD = 6.11$) and females ($M = 19.5, SD = 5.58$) in the NC subscale, $t(352) = 0.55, p = .580, d = 0.049$.

As seen in Table 4, female participants in our study also exhibited significantly greater eco-guilt ($M = 30.9, SD = 5.98$) compared to males ($M = 29.5, SD = 5.94$), $t(383) = -2.82, p = .005$. This was observed as well in terms of eco-grief, females experienced significantly greater eco-grief ($M = 18.7, SD = 2.85$) compared to males ($M = 17.7, SD = 3.10$), $t(353) = -3.85, p < .001$ in the sample.

When grouped according to place of residence a Welch's t-test was also used to account of the unequal residence distribution in the sample (463 urban and 153 rural). Our sample of Gen Z Filipinos results in Table 5 revealed no significant difference in the Habitual Ecological Worry eco-anxiety subscale scores between urban ($M = 44.5, SD = 5.78$) and rural residents ($M = 43.9, SD = 6.28$), $t(243) = 1.05, p = .296, d = 0.100$. However, in the eco-anxiety Negative Consequences subscale, rural residents scored significantly higher ($M = 21.5, SD = 5.63$) than urban residents ($M = 19.0, SD = 5.66$), $t(261) = -4.67, p < .001$.

Table 5 also presents no significant difference between urban ($M = 30.4, SD = 6.10$) and rural residents ($M = 30.6, SD = 5.70$), in terms of eco-guilt, $t(276) = -0.269, p = .789$. The same is found as regards with the feeling of eco-grief, there was no significant difference, $t(280) = -1.08, p = .280$, between urban ($M = 18.3, SD = 3.03$) and rural residents ($M = 18.6, SD = 2.79$).

While generally expressing worry, guilt, and grief about the environment, Filipino Gen Z participants in Study 1 did not report high levels of anxiety regarding the negative consequences of the ecological crisis. However, compared to males, female participants have higher habitual ecological worry, ecological guilt, and ecological grief. Both male and female Gen Zs also experience the same degree of anxiety in terms of the negative consequences of environmental crisis. In terms of place of residence, quantitative results showed that in all scales, Gen Z participants in rural residences are more anxious regarding negative consequences as compared to urban residents.

After completing the quantitative investigation of the study, we moved on to the qualitative study to explore more deeply how Gen Z Filipinos understand and navigate their experiences of eco-emotions in relation to both their gender and residential context. The interview central question: *How do Gen Z Filipinos experience and make sense of their eco-anxiety, eco-grief, and eco-guilt in the context of their gender and place of residence?* in our qualitative Study 2 was then derived to sequentially explain these quantitative results.

Table 4. Differences in Eco-Anxiety, Eco-Guilt, and Eco-Grief Levels Between Males and Females (N = 616)

Ecological emotions	Males (n = 196)		Females (n = 420)		t	p	Cohen's d
	M	SD	M	SD			
Eco-anxiety (HEW)	43.1	6.16	44.9	5.71	-3.41	<.001	0.299
Eco-anxiety (NC)	19.8	6.11	19.5	5.58	0.55	.580	0.049
Eco-guilt	29.5	5.94	30.9	5.98	-2.82	.005	0.243
Eco-grief	17.7	3.10	18.7	2.85	-3.85	<.001	0.338

Note. Significant p values appear in bold.

Table 5. Differences in Eco-Anxiety, Eco-Guilt, and Eco-Grief Levels Between Urban and Rural Residents (N = 616)

Ecological emotions	Urban (463 urban residents)		Rural (153 rural residents)		<i>t</i>	<i>p</i>	Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			
Eco-anxiety (HEW)	44.5	5.78	43.9	6.28	1.05	.296	0.100
Eco-anxiety (NC)	19.0	5.66	21.5	5.63	-4.67	<.001	0.435
Eco-guilt	30.4	6.10	30.6	5.70	-0.269	.789	0.025
Eco-grief	18.3	3.03	18.6	2.79	-1.08	.280	0.099

Note. Significant *p* values appear in bold.

Study 2: Qualitative

Nine participants provided informed consent to participate in Study 2. Of these, five resided in urban areas and four in rural areas; the sample comprised five males and four females. A gender imbalance was observed in Study 1, with females disproportionately represented in the quantitative sample. This imbalance was considered during the selection of participants for Study 2, and deliberate efforts were made to include representation from both genders among those who expressed willingness to participate in qualitative interviews. As a result, Study 2 achieved a nearly equal distribution of male and female participants, which served to partially mitigate the female-heavy composition of the initial sample. Although quantitative findings indicated higher levels of ecological worry, guilt, and grief among female participants, the qualitative analysis was designed to examine whether the nature and expression of these ecological emotions differed across genders, thereby ensuring that male participants' perspectives were adequately captured and interpreted.

During the interviews, participants discussed how gender and place of residence shaped their emotional experiences and responses to the ecological crisis. The "ecological crisis," while broad, serves as the unifying phenomenon through which our diverse participants navigate their emotions. Our goal was not to find a single experience but to see how the same crisis is "socially lived and emotionally shared" across different settings.

Participants across different gender identities and residential contexts generally expressed similar thoughts and feelings. This consistency reinforced the shared nature of their experiences and confirmed the decision not to create separate thematic distinctions for these non-significant variables. The focus was therefore directed toward themes that meaningfully expanded upon the study's statistically significant findings.

To present a more cohesive narrative, the findings are organized by overarching themes rather than being separated strictly by gender or location. This approach emerged from the analysis, which revealed that themes were consistent across these demographics, often expressed through shared experiences or participants' observations about others. While minor demographic patterns did surface, they are woven into the broader discussion of the main themes to maintain clarity and present a unified picture of how Gen Z Filipinos experience and express ecological emotions. Moreover, the findings of the qualitative sample, drawn from the personal narratives of the selected participants, are interpreted as patterns, tendencies, or suggestibility.

Thematic Findings Based on Gender Differences

From the qualitative data, three central themes emerged related to how Gen Z participants experience eco-emotions through the lens of gender. These themes provide depth to the statistically significant findings from the quantitative study, particularly where female participants scored significantly higher in habitual ecological worry, eco-grief, and eco-guilt compared to males.

Theme 1: Gendered Emotional Expression

This theme highlights the perceived differences in how ecological emotions—particularly eco-anxiety, eco-grief, and eco-guilt—are expressed or withheld depending on gender. Female participants frequently described being more open and emotionally expressive, citing crying, verbalizing their concerns, or actively discussing environmental issues with peers. They felt it was acceptable—if not expected—for women to express such emotions. As one female participant explained:

“I think those who usually care more are the moms or, generally, girls. As a girl, I tend to worry easily about my surroundings, about nature, and where I am. I feel like girls are more likely to express what they feel. There are men who are expressive too, but I think they’re more likely to keep it to themselves and ask themselves first, ‘Do I really need to share this?’” (**Participant 9, Female, 21, Rural**)

Another participant echoed a similar view, emphasizing how women’s emotionality contributes to their heightened ecological concern:

“Women are more emotional than men, and I think women care more about the environment. Because they’re more emotional, they tend to think more about environmental issues than men. It’s not that men don’t care, but they don’t really think much about those kinds of things. Men are usually more focused on other matters.” (**Participant 4, Female, 21, Rural**)

Interestingly, male participants shared similar observations. Some noted that although men also feel concern or frustration, they often express it differently—less through overt emotion and more through acknowledgment or discussion in neutral tones. As male participant echoed this, noting:

“I think it’s mostly the females who are more aware and genuinely concerned about these issues—like, they really care. In my experience, women are also more vocal when expressing their emotions. When my female friends talk about these things, I can really feel their anger or frustration. But with my male friends, it’s more like they agree or acknowledge that something is wrong, but it’s rare for me to actually hear or feel any strong emotion from them.” (**Participant 6, Male, 22, Urban**)

Theme 2: Responsibility Rooted in Nurturing Roles

Several female participants associated their ecological concern with a sense of duty rooted in traditional nurturing roles. They felt accountable not just for themselves, but also for how ecological harm might affect others—particularly family members, future children, or younger generations. This perceived responsibility seemed to deepen their feelings of eco-guilt and ecological worry. As one participant explained:

“Uhm... ah, I think with men it’s more rare. For example, my female friends tend to care more about things like segregating trash in school. Men, on the other hand, often just throw things wherever there’s space—they don’t really think about whether it’s plastic and should go somewhere else.” (**Participant 4, Female, 21, Rural**)

Beyond immediate caregiving, some women also linked their sense of responsibility to potential future roles as mothers. They reflected on how their ecological awareness shaped the way they thought about childbearing and the world their children might inherit. This anticipation of future responsibility extended their ecological emotions toward imagined generations. As one participant shared:

“I guess being a woman has some factor to it, since as a woman, it’s up to my choice if I choose to have children in the future. I feel like if I decide to have children, I would also have to consider factors like the future environmental state of my future children. I feel like there aren’t a lot of men who would consider those as a factor. They tend to have other priorities—maybe more economic ones. That’s what they would consider rather than these.” (**Participant 5, Female, 21, Urban**)

This sense of accountability—both present and future—illustrates how ecological emotions among women were shaped by socially reinforced caregiving expectations. For many, nurturing extended beyond people to encompass care for the environment itself, transforming what might otherwise be a personal concern into a moral and relational duty.

Theme 3: Emotional Resonance and Collective Grief

This theme captures how Gen Z participants were not only aware of their own ecological emotions but also deeply affected by the emotional reactions of others—particularly those within their social and peer networks. Unlike Theme 1, which focused on how individuals regulate and express their eco-emotions based on gender norms, this theme explores how emotions are socially transmitted, especially through moments of collective expression and shared distress.

Participants described instances where their friends’ or peers’ emotional responses—especially those of female friends—amplified their own awareness or concern. This phenomenon of “catching” emotions, also known as emotional contagion,

shaped their experience of the ecological crisis as a shared emotional burden, not merely a personal one. One male participant reflected on how his emotional landscape was shaped by those around him:

“I think being a woman plays a role here, since I’ve noticed that women tend to be more emotionally expressive. And when I see my female friends talk about these issues, I can really sense how affected they are. I end up feeling it too—sometimes their emotions rub off on me.” **(Participant 7, Male, 21, Rural)**

Another participant echoed this idea of collective awareness and empathy:

“When they talk about it passionately—especially my girlfriends—you can really feel the sadness or anger. Even if I wasn’t feeling it strongly before, it starts to hit me. It is like a group feeling.” **(Participant 4, Female, 21, Rural)**

In these narratives, the ecological crisis became more than a personal concern—it became socially lived and emotionally shared. The grief of one participant often evoked a ripple effect, deepening others’ sense of urgency, worry, or sadness. This underscores the role of peer dynamics and group sentiment in shaping emotional responses to environmental degradation.

Thematic Findings Based on Residence Differences

Quantitative findings revealed that rural participants scored significantly higher than urban participants on the negative consequences subscale of eco-anxiety. This pattern was mirrored in the qualitative data, which yielded three key themes based on participants’ place of residence:

Theme 4: Proximity to Environmental Harm

These results are confined to tendencies evident in participant narratives and should not be interpreted as generalizable to all rural Filipinos, indicating only a possible association between proximity to environmental harm and emotional impact within the sample.

Rural participants consistently described witnessing environmental degradation firsthand. Rather than viewing the ecological crisis as distant or abstract, they spoke of polluted rivers they used to swim in, deforestation in their hometowns, and agricultural decline experienced by their families. These direct encounters made the threats of climate change feel more personal and urgent.

“Maybe yes, because I’ve been living in a rural area, and here I really see a lot of trees and things like that. So, I think my living in a rural place affects it, because there’s a big difference between rural and urban areas. The area here in Quezon, for example, isn’t really that urbanized because the soil here is still good for farming... It’s scary and worrisome, especially for people living in places that are more prone to flooding. I remember I had a classmate before who lived in Malvar (flood-prone area).” **(Participant 4, Female, 21, Rural)**

In contrast, urban participants often mentioned daily environmental stressors such as traffic, pollution, flooding, or extreme heat. While these narratives conveyed frustration and concern, they primarily framed environmental decline as part of daily inconvenience or diminished livability. Their concerns reflected exposure to harm, but not necessarily a deeper sense of personal or cultural loss. As one participant explained:

“I feel like... being in an urban area does affect these worries. I often see news about flooding—like, even just a bit of rain and it already floods. When you go to Manila, you really have to be cautious about those things. And it also gets really hot quickly, since there aren’t many trees around. But when you go to rural areas, it feels more breathable—it’s still hot, but there’s more air, you can really feel it. I remember Manila Bay used to be beautiful, but now there are so many buildings and even artificial islands. It doesn’t look as nice anymore—it’s mostly infrastructure now.” **(Participant 5, Female, 21, Urban)**

Theme 5: Disruption of Tradition and Place Identity

Urban participants described the emotional toll of witnessing once-familiar landscapes undergo rapid transformation due to modernization and environmental degradation. These accounts revealed how the ecological crisis intersects with a deeper sense of place-based grief—a sadness that stems not only from environmental harm, but from the loss of memories, routines,

and cultural touchstones associated with those places. The narrative tone was one of nostalgia, mourning, and emotional dislocation.

Participants spoke about green spaces, rice fields, and open areas that were once central to their everyday lives, particularly during childhood, but have since been replaced by infrastructure, traffic, and commercial development. These changes appeared to evoke feelings of sadness, disconnection, and a growing sense that the places they once called home no longer reflected their identity or sense of belonging.

“Since new establishments were built, our area became more modernized... the places where we used to play when we were kids—like rice fields—they’re all gone. Now, it’s always traffic. It makes me sad, like the Marikina I experienced when I was younger is no longer there.” **(Participant 6, Male, 22, Urban)**

“I remember Manila Bay used to be beautiful, but now there are so many buildings... you can see it’s not really that beautiful anymore because it’s all infrastructure now.” **(Participant 5, Female, 21, Urban)**

In contrast to the rural narratives, which focused on environmental disruption as a threat to survival or livelihood, urban participants often framed their eco-anxiety through the lens of identity loss and emotional detachment. The negative consequences of environmental decline were experienced not only in terms of physical discomfort (e.g., heat, flooding), but also as the erosion of continuity between past and present—of witnessing their city shift away from what it once meant to them.

Discussion

Guided by a mixed-methods sequential explanatory (QUAN → qual) design, the present discussion brings together the quantitative patterns identified in Study 1 and the qualitative narratives explored in Study 2 to illuminate how Generation Z Filipinos experience and make sense of eco-anxiety, eco-guilt, and eco-grief.

Study 1

This study investigated eco-anxiety, eco-guilt, and eco-grief among Gen Z Filipinos, and examined how these ecological emotions vary by gender and place of residence. Our quantitative findings revealed that participants exhibited high levels of habitual ecological worry, eco-guilt, and eco-grief, yet reported relatively low negative functional consequences. Notably, females demonstrated significantly higher habitual ecological worry, eco-guilt, and eco-grief than males, though both genders reported similar levels of functional impairment. Rural residents experienced greater negative consequences from eco-anxiety than urban residents, despite comparable levels of ecological worry, guilt, and grief across residential contexts.

Study 1 findings indicate that Gen Z Filipinos experience elevated levels of affective ecological distress, characterized by high habitual worry, guilt, and grief that exceed normative midpoints. However, the relatively lower scores on the negative consequences subscale suggest that while these youth are deeply preoccupied with the ecological crisis, their distress may manifest more as a chronic internal state than as an acute expectation of immediate personal disaster. This elevated emotional burden was particularly pronounced among female participants, who consistently reported greater worry, guilt, and grief than their male counterparts—a trend that aligns with broader psychological literature on gendered responses to environmental threats.³²

Geographically, while place of residence did not influence general affective worry or grief, the heightened anxiety regarding negative consequences among rural residents suggests that proximity to environmental degradation and climate vulnerability may shape specific dimensions of ecological anxiety.⁸ Taken together, these results highlight a generation that is emotionally attuned to global ecological decline yet influenced by demographic and locational factors in how they perceive personal risk.²⁸

Study 2

In the quantitative study, no statistically significant differences were found for certain ecological emotions, specifically eco-anxiety across gender, and eco-grief and eco-guilt across place of residence. The subsequent qualitative data corroborated this, as no strong or distinct patterns emerged to justify separate themes for these variables. The qualitative analysis of interview data yielded five themes that help illuminate these patterns: *Gendered Emotional Expression, Responsibility Rooted in Nurturing Roles, Emotional Resonance and Collective Grief, Proximity to Environmental Harm, and Disruption of Tradition and Place Identity.*

Study 2 qualitative findings expound the nuanced, lived experiences of Gen Z's eco-emotions, providing a socio-cultural context for the observed gender and residential disparities. Gendered differences are characterized by a higher emotional expressivity among women, who frame their ecological worry through the lens of socially reinforced nurturing roles and a perceived moral duty to future generations⁶⁸; conversely, men tend to adopt more neutral, detached tones despite sharing similar underlying concerns. This emotional landscape is further intensified by emotional contagion, where collective grief within peer networks transforms personal anxiety into a shared social burden.⁸

Regarding residence, the qualitative data reveals a dichotomy between proximity and identity: rural participants experience eco-anxiety as an urgent, personal threat tied to the tangible degradation of livelihoods and natural resources,³⁶ whereas urban participants process environmental decline through place-based grief. For urbanites, the transformation of familiar landscapes into industrialized spaces triggers a sense of nostalgia and "solastalgia," where the negative consequences of the ecological crisis are felt as an erosion of cultural identity and a loss of connection to their personal history.^{69,70}

Gendered Emotional Expression

Both quantitative results and qualitative accounts indicated that females express ecological emotions more openly than males. Participants perceived women as more proactive and vocal about environmental concerns, while men tended to acknowledge issues in more reserved, neutral tones. These findings align with research demonstrating greater environmental concern and willingness to express such emotions among women.^{43,44,48} However, a critical interpretive question remains about whether these differences reflect genuinely distinct emotional experiences or socialized patterns of emotional expression. Gender role socialization may encourage women to articulate environmental distress while discouraging equivalent expression among men.⁷¹ The finding that both genders reported similar functional consequences suggests that underlying stress responses may be more comparable than overt expression implies.

Responsibility Rooted in Nurturing Roles

Female participants linked their ecological concern to caregiving responsibilities extending to future generations. Several expressed uncertainties about childbearing given environmental degradation, which is consistent with findings from existing research on eco-reproductive concerns.^{72,73} This pattern reflects how the socialization of women into caregiving roles may amplify ecological emotions by expanding the perceived scope of responsibility beyond the self to encompass families, communities, and unborn generations.^{45,74} The extension of nurturing tendencies to encompass nature itself⁴⁴ may partially explain the elevated levels of eco-guilt and eco-grief observed among women in this study.

Emotional Resonance and Collective Grief

Participants described how others' ecological emotions influenced their own, with expressions of grief creating ripple effects that amplified collective concern. This emotional contagion phenomenon⁷⁵ suggests that ecological emotions are not merely individual psychological states but socially transmitted experiences. The collective nature of eco-grief has important implications for both stress propagation and potential intervention: while shared distress may intensify anxiety, it may also foster solidarity and motivate collective climate action.⁷⁶ Cunsolo and Landman similarly documented how actively listening to others' environmental losses can evoke eco-grief even without direct personal experience—a finding our participants corroborated.⁷⁷

Proximity to Environmental Harm

Elevated negative consequences from eco-anxiety among rural dwellers in our study reflect their direct exposure to environmental degradation. Unlike urban residents, whose environmental stressors (heat, pollution, flooding) are often framed as daily inconveniences, rural communities experience ecological change as an existential threat to livelihoods dependent on natural resources.^{24,46} The qualitative findings provide further context to this pattern, illustrating how rural participants encounter environmental degradation firsthand, making ecological threats more immediate and emotionally impactful. Meanwhile, urban participants tend to frame environmental concerns in relation to daily stressors such as heat, flooding and pollution. This distinction between abstract concern and concrete impact may explain why rural and urban residents report similar worry levels yet differ in functional consequences. The finding aligns with research indicating that proximity to environmental harm intensifies the psychological burden of ecological crisis.^{5,76} However, the findings from the

personal experiences of the selected interview participants should be interpreted as context-specific to our sample; as such, the analysis is intended to highlight how proximity to environmental harm is experienced within a particular setting.

Disruption of Tradition, Place Identity Loss, and Emotional Dislocation

Urban participants articulated a distinctive form of ecological grief centered on place-based identity loss. Narratives of childhood landscapes transformed by development evoked nostalgia and disconnection; what Benham and Hoerst¹⁶ describe as unsettled place attachment. This place-based grief differs qualitatively from rural concerns about resource depletion; it represents the erosion of continuity between past and present selves. The ‘extinction of experience’ accompanying urbanization⁷⁸ may thus generate eco-anxiety rooted in identity disruption rather than immediate survival concerns.

The ecological crisis has profoundly disrupted the traditional Filipino connection to the land, transforming cherished landscapes into sources of deep emotional distress. For many Gen Z Filipinos in our study, this environmental degradation triggers a sense of *pagpanumdom*—a melancholic remembrance/nostalgia for a past that is being rapidly erased (e.g., “*Mourning the Marikina I experienced when I was younger.*”). This is most poignantly seen in the disappearance of *bukid* (rice fields) and open green spaces that once served as central cultural touchstones for childhood play and community bonding (e.g., “*The places where we used to play... like rice fields... they’re all gone.*”). As these natural areas are replaced by industrialization and modernization, individuals experience an erosion of *panunuluyan* (a sense of belonging), leading to a “place-based grief” where their home no longer reflects their identity (e.g., “*Feelings that home no longer reflected their identity or sense of belonging.*”). This “extinction of experience” represents a loss of continuity between their past and present selves, as the environmental stability essential for their psychological well-being is replaced by a lingering ontological insecurity.

Theoretical Integration

Giddens’ 1991 theory of ontological security provides a unifying framework for these findings. Ecological emotions may arise from disruptions to the sense of environmental stability essential for psychological wellbeing.⁷⁹ The continuous environmental losses documented by our participants, whether degraded rural landscapes or transformed urban spaces, appear to induce ontological insecurity, which manifests as anxiety, guilt, and grief.⁷ While the quantitative scales used to measure these emotions were not formally adapted, this mixed-methods approach provides the necessary cultural grounding by using qualitative findings to interpret these standardized constructs. Specifically, the high reliability of the EAQ-22, EGuiQ-11, and EGriQ-6 suggests that these theoretically grounded constructs resonate with the experiences of Gen Z Filipinos. This psychometric evidence, contextualized through indigenous concepts like *pagpanumdom* (remembrance/nostalgia) and *panunuluyan* (belonging), ensures that the study’s conceptual coherence is rooted in the specific lived realities of the Global South. Critically, this framework suggests that ecological emotions, while distressing, may also motivate adaptive responses as individuals seek to restore security through environmental engagement and community action.

Our findings revealed a dissociation between high ecological worry and low functional impairment warrants further investigation. While this dissociation is noteworthy, our study cannot definitively explain its occurrence. This pattern may reflect effective coping mechanisms, strong social support networks, or cultural factors our participants that buffer against stress-related dysfunction. Alternatively, it may indicate that current measures inadequately capture culturally specific expressions of ecological distress. Understanding why some individuals maintain functioning despite significant ecological concern has implications for identifying protective factors and developing targeted interventions. These cultural explanations are speculative, while our data confirms the “high worry/low impairment” pattern, the reasons (such as cultural resilience) are inferred trends that require further empirical investigation.

Triangulation

By integrating Study 1 and Study 2 we reveal a complex emotional landscape among Gen Z Filipinos, where statistical trends are deeply contextualized by personal and cultural narratives. Quantitatively, Study 1 identifies a unique “dissociation” where our participants report high levels of habitual ecological worry, guilt, and grief that exceed normative midpoints, yet maintain relatively low functional impairment. Qualitative themes from Study 2 explain these disparities through social and geographical lenses: for instance, while Study 1 found that females report significantly higher levels of emotional distress, Study 2 suggests this is driven by Gendered Emotional Expression and Responsibility Rooted in Nurturing Roles, as women often frame ecological concern through a moral duty to future generations.

Geographically, although both groups share similar levels of general worry, rural residents in our sample significantly reported higher negative consequences from eco-anxiety, which the theme Proximity to Environmental Harm attributes to

their direct, survival-based exposure to agricultural decline and deforestation. Conversely, urban participants navigate a distinctive place-based grief characterized by *pagpanumdom* (remembrance/nostalgia) for lost childhood landscapes like *bukid* (rice fields), experiencing the crisis as an erosion of cultural identity and *panunuluyan* (belonging) rather than an immediate physical threat.

Together, these studies, as depicted in Figure 1, illustrate that ecological distress in the Philippines is a socially lived phenomenon where emotional vulnerability is dictated by one's specific socio-ecological position.

Limitations and Future Directions

Several limitations warrant consideration. First, sample size, as the overrepresentation of females and urban residents in our investigations may bias findings. While the present study aimed to examine gender differences, the disproportionate number of female participants compared to male participants represents a methodological constraint. This imbalance potentially introduces a self-selection bias, as individuals with higher baseline ecological distress—disproportionately women in this sample—may have been more motivated to participate. While the sample size for men remained sufficient for certain comparative analyses, the disparity may have influenced the variance and the overall representativeness of the emotional profiles generated. Consequently, the results should be interpreted with caution when applied to male-identifying individuals. Future research should employ purposive sampling strategies to achieve a more equitable gender distribution and verify if these emotional patterns hold across more balanced cohorts. The Study 2 sample of nine participants, while within qualitative research guidelines, constrained our ability to draw robust comparisons across gender and residential categories. Future studies should ensure balanced representation and larger qualitative samples to enable meaningful subgroup analyses.

Second, voluntary response sampling introduces self-selection bias; individuals with stronger ecological concerns may have been more motivated to participate. Third, the cross-sectional design captures eco-emotions at a single timepoint, precluding examination of how these emotions develop or fluctuate in response to environmental events. Longitudinal designs would better illuminate the temporal dynamics of ecological emotions and their relationship to ongoing crises.

Fourth, the Philippine-specific context limits generalizability. While this specificity is also a strength: contributing Global South perspectives to a predominantly WEIRD literature, future research should extend to other climate-vulnerable nations to identify shared versus culturally distinct patterns. Fifth, our binary treatment of gender excludes LGBTQIA+ perspectives, despite evidence that these populations face heightened climate vulnerability.⁸⁰ Future research should incorporate diverse gender identities.

Lastly, our study concerns the use of the original English versions of the instruments without formal cultural adaptation or pilot testing within the local context. While English is the primary medium of instruction and the official language for

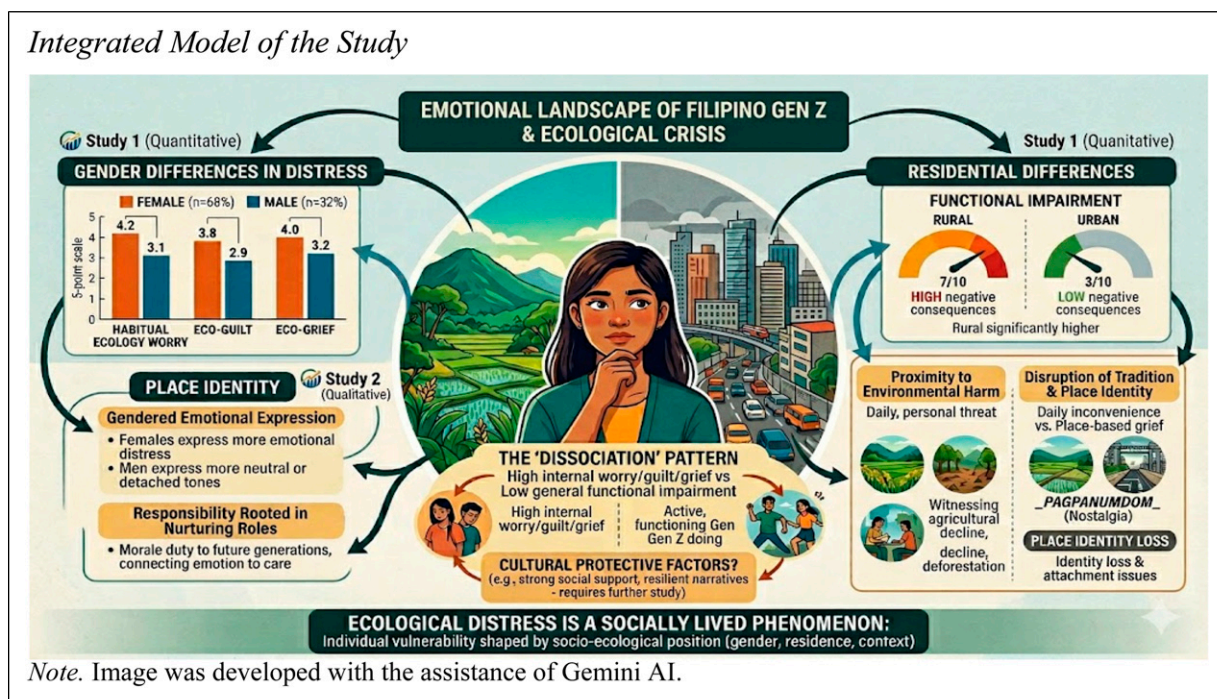


Figure 1. Integrated model of the study

professional and academic assessments in the Philippines, the absence of a rigorous translation-back-translation process may have overlooked specific cultural nuances or idiomatic expressions unique to the Filipino population. Although the inclusion criterion of a ‘fair understanding of English’ was implemented to ensure comprehension and all quantitative measures show reasonable reliability coefficients in the present study, self-reported proficiency may not fully account for subtle linguistic barriers that could influence item interpretation. Specifically, the strong internal consistency of the EAQ-22, EGuiQ-11, and EGriQ-6 suggests that these theoretically grounded constructs resonate with the experiences of Gen Z Filipinos. This psychometric evidence, combined with the high level of English proficiency among the participants, provides a baseline level of validation that supports the manuscript’s claims regarding the manifestation of ecological emotions in a climate-vulnerable Global South nation. As such, to enhance the psychometric robustness of these findings, future research should prioritize the cultural and linguistic validation of these scales. This includes conducting cognitive interviews and pilot testing to ensure the instruments are culturally resonant. Subsequent studies might also benefit from developing and validating versions in Filipino or other local languages to compare responses across different linguistic demographics within the Philippines, thereby ensuring broader inclusivity and cross-cultural equivalence.

Several promising directions emerge from our findings. The role of socioeconomic status, raised by participants themselves, merits investigation given evidence that lower SES increases vulnerability to environmental harms.⁸¹ Intergenerational comparisons could illuminate whether Gen Z’s ecological emotions reflect developmental stage, historical context, or both. In addition, participants’ accounts also offered preliminary indications that eco-emotions may be experienced and shared within peer networks, suggesting that social network approaches may be useful for further exploring how ecological distress may circulate among individuals and how peer support might be leveraged for intervention. Finally, the apparent disconnect between high worry and low functional impairment invites investigation of protective factors that may potentially inform resilience-focused interventions.

Conclusion and Implications

This study provides preliminary empirical insight into ecological emotions among a climate-vulnerable Global South population, contributing to a growing but still limited body of research in this area. By offering initial evidence from a climate-vulnerable Global South context, these findings underscore the need for expanded empirical investigation and context-sensitive approaches to understanding ecological emotions.

Gen Z Filipinos experience substantial eco-anxiety, eco-guilt, and eco-grief, with important variations by gender and residential context. Females report higher levels of habitual ecological worry, guilt, and grief. These patterns are likely shaped by gendered socialization into caregiving roles and differential norms around emotional expression. Nevertheless, while women may be more emotionally expressive, concerns about the ecological crisis had similarly low levels of impact on psychological functioning across both genders. Rural residents experience greater functional consequences from eco-anxiety, reflecting direct dependence on threatened natural resources. Urban residents, meanwhile, articulate place-based grief tied to the psychological consequences of rapid urbanization, including the loss of green spaces, increased pollution, and the replacement of natural habitats.

Drawing from the theory of Ontological Security or Insecurity, our findings suggest that Gen Z Filipinos’ ecological emotions arise from disruptions in their sense of security within their surroundings. The continuous environmental loss further induces ontological insecurity, which manifests as eco-anxiety, eco-guilt, and eco-grief. However, these same emotions may also motivate individuals to seek other ways of security through environmental care and community engagement.

For mental health practitioners, these findings underscore the importance of recognizing ecological emotions as legitimate psychological experiences warranting clinical attention particularly among young people in climate-vulnerable regions. The low negative psychological consequences of eco-anxiety despite high levels of worry suggest that most Gen Z Filipinos are coping adaptively, but vigilance is warranted for those whose functioning deteriorates. For educators and environmental programs, acknowledging the emotional dimensions of ecological crisis rather than focusing solely on information provision may enhance engagement and wellbeing. Group-based interventions that leverage the collective nature of eco-emotions may be particularly effective.

By situating eco-anxiety, eco-guilt, and eco-grief within socio-cultural and place-based contexts, this study contributes to an emerging literature recognizing that ecological emotions are shaped by culture, community, and geography. Understanding these contextual influences is essential for developing interventions that resonate with the lived experiences of those most affected by environmental crisis.

ORCID iD

Marc Eric S. Reyes  <https://orcid.org/0000-0002-5280-1315>

Ethical Considerations

All procedures performed in the present study that involved human participants were per the ethical standards from Davao de Oro State College – Research Ethics Committee with Protocol Code: 824-08-2024.

Consent to Participate

The current study gave informed consent before voluntary participation. In addition, participants were briefed on the nature of the study, assured that all data collected would be kept confidential, and that participation was voluntary without remuneration.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The University of Santo Tomas Research Center for Social Sciences and Education supported Marc Eric S. Reyes, Ma. Criselda T. Pacquing, Karen Katrina V. Trinidad.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Available upon request. Consent was not sought from participants that the data will be made available publicly.

Use of Generative AI

Generative AI tools, including Grammarly, Gemini AI, and Microsoft Copilot, were used only for language editing, clarity, and formatting/image support. AI tools were not used in the conception of the study, data analysis, or interpretation of results. The manuscript reflects the authors' original work.

References

- Aharkov O, Prisniakova L, Pashko A, Tarasevych I. Psychological consequences of environmental crises for the individual. *E3S Web of Conferences*. 2023;452:07016. doi:[10.1051/e3sconf/202345207016](https://doi.org/10.1051/e3sconf/202345207016).
- Nilsen HR. Code red for humanity: The role of business ethics as we transgress planetary thresholds. *Journal of Business Ethics*. 2023; 189(3):1-7. doi:[10.1007/s10551-023-05402-3](https://doi.org/10.1007/s10551-023-05402-3).
- Stanley SK, Hogg TL, Leviston Z, Walker I. From anger to action: Differential impacts of eco-anxiety, eco-depression, and eco-anger on climate action and wellbeing. *The Journal of Climate Change and Health*. 2021;1:100003. doi:[10.1016/j.joclim.2021.100003](https://doi.org/10.1016/j.joclim.2021.100003).
- Kalwak W, Weihgold V. The Relationality of Ecological Emotions: An interdisciplinary critique of individual resilience as Psychology's response to the Climate Crisis. *Frontiers in Psychology*. 2022;13:823620. doi:[10.3389/fpsyg.2022.823620](https://doi.org/10.3389/fpsyg.2022.823620).
- Ágoston C, Urban R, Nagy B, et al. The psychological consequences of the ecological crisis: Three new questionnaires to assess eco-anxiety, eco-guilt, and ecological grief. *Climate Risk Management*. 2022;37(1):100441. doi:[10.1016/j.crm.2022.100441](https://doi.org/10.1016/j.crm.2022.100441).
- Cunsolo A, Ellis NR. Ecological grief as a mental health response to climate change-related loss. *Nature Climate Change*. 2018;8(4): 275-281. doi:[10.1038/s41558-018-0092-2](https://doi.org/10.1038/s41558-018-0092-2).
- Pihkala P. Anxiety and the ecological crisis: An analysis of eco-anxiety and climate anxiety. *Sustainability*. 2020;12(19):7836. doi:[10.3390/su12197836](https://doi.org/10.3390/su12197836).
- Hickman C, Marks E, Pihkala P, et al. Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey. *The Lancet Planetary Health*. 2021;5(12):e863-e873. doi:[10.1016/s2542-5196\(21\)00278-3](https://doi.org/10.1016/s2542-5196(21)00278-3).
- Ágoston C, Csaba B, Nagy B, et al. Identifying Types of Eco-Anxiety, Eco-Guilt, Eco-Grief, and Eco-Coping in a Climate-Sensitive Population: A Qualitative Study. *International Journal of Environmental Research and Public Health*. 2022;19(4):2461. doi:[10.3390/ijerph19042461](https://doi.org/10.3390/ijerph19042461).
- Brosch T. Affect and emotions as drivers of climate change perception and action: a review. *Current Opinion in Behavioral Sciences*. 2021;42:15-21. doi:[10.1016/j.cobeha.2021.02.001](https://doi.org/10.1016/j.cobeha.2021.02.001).
- Hickman C. We need to (find a way to) talk about ... Eco-anxiety. *Journal of Social Work Practice*. 2020;34(4):411-424. doi:[10.1080/02650533.2020.1844166](https://doi.org/10.1080/02650533.2020.1844166).
- Tsevreñi I, Proutsos N, Tsevreñi M, Tigkas D. Generation Z Worries, Suffers and Acts against Climate Crisis—The Potential of Sensing Children's and Young People's Eco-Anxiety: A Critical Analysis Based on an Integrative Review. *Climate*. 2023;11(8):171. doi:[10.3390/cli11080171](https://doi.org/10.3390/cli11080171).
- Pihkala P. Eco-anxiety and environmental education. *Sustainability*. 2020;12(23):10149. doi:[10.3390/su122310149](https://doi.org/10.3390/su122310149).

14. Nambiar V, Singh S. Understanding The Prevalence of Eco Guilt and Eco Grief among Adolescents: A Salutogenic Approach Towards Building “Pro-Environmental Behaviour.”. *IOP Conference Series: Earth and Environmental Science*. 2023;1279:012019. doi:[10.1088/1755-1315/1279/1/012019](https://doi.org/10.1088/1755-1315/1279/1/012019).
15. Moore MM, Yang J. Using Eco-Guilt to Motivate Environmental Behavior Change. *Environmental Communication*. 2020;14(4):522-536. doi:[10.1080/17524032.2019.1692889](https://doi.org/10.1080/17524032.2019.1692889).
16. Benham C, Hoerst D. What role do social-ecological factors play in ecological grief?: Insights from a global scoping review. *Journal of Environmental Psychology*. 2024;93:102184. doi:[10.1016/j.jenvp.2023.102184](https://doi.org/10.1016/j.jenvp.2023.102184).
17. Rai A. Understanding Eco-Anxiety and Eco-Grief: A Comprehensive Reviews. *International Journal of Research Publication and Reviews*. 2023;4(8):681-686. doi:[10.55248/gengpi.4.823.50353](https://doi.org/10.55248/gengpi.4.823.50353).
18. Voşki A, Wong-Parodi G, Ardoin NM. A new planetary affective science framework for eco-emotions: findings on eco-anger, eco-grief, and eco-anxiety. *Global Environmental Psychology*. 2023;1:1-31. doi:[10.5964/gep.11465](https://doi.org/10.5964/gep.11465).
19. Stafford AM, Walton AL, Gonzalez-Guarda RM. Growing up in an era of storms and stress—Promoting hope among adolescents in the face of climate change. *JAMA Health Forum*. 2023;4(9):e233834. doi:[10.1001/jamahealthforum.2023.3834](https://doi.org/10.1001/jamahealthforum.2023.3834).
20. Ogunbode CA, Pallesen S, Böhm G, et al. Negative emotions about climate change are related to insomnia symptoms and mental health: Cross-sectional evidence from 25 countries. *Current Psychology*. 2021;42:845-854. doi:[10.1007/s12144-021-01385-4](https://doi.org/10.1007/s12144-021-01385-4).
21. Kleres J, Wettergren Å. Fear, hope, anger, and guilt in climate activism. *Social Movement Studies*. 2017;16(5):507-519. doi:[10.1080/14742837.2017.1344546](https://doi.org/10.1080/14742837.2017.1344546).
22. Garcia MEJ, Hernandez AA. Pattern Analysis of Natural Disasters in the Philippines. *Big Data Technologies and Applications*. 2018;248:74-83. doi:[10.1007/978-3-319-98752-1_9](https://doi.org/10.1007/978-3-319-98752-1_9).
23. Bollentino V, Alcayna-Stevens T, Sharma M, Dy P, Pham P, Vinck P. Public perception of climate change and disaster preparedness: Evidence from the Philippines. *Climate Risk Management*. 2020;30:100250. doi:[10.1016/j.crm.2020.100250](https://doi.org/10.1016/j.crm.2020.100250).
24. Aruta JJBR, Guinto RR. Climate anxiety in the Philippines: Current situation, potential pathways, and ways forward. *The Journal of Climate Change and Health*. 2022;6:100138. doi:[10.1016/j.joclim.2022.100138](https://doi.org/10.1016/j.joclim.2022.100138).
25. Santos GDC. 2020 tropical cyclones in the Philippines: A review. *Tropical Cyclone Research and Review*. 2021;10(3):191-199. doi:[10.1016/j.tcr.2021.09.003](https://doi.org/10.1016/j.tcr.2021.09.003).
26. Cianconi P, Hanife B, Grillo F, Lesmana CBJ, Janiri L. Eco-emotions and Psychoterratic Syndromes: Reshaping Mental Health Assessment Under Climate Change. *The Yale Journal of Biology and Medicine*. 2023;96(2):211-226. doi:[10.59249/EARX2427211](https://doi.org/10.59249/EARX2427211).
27. Ojala M, Cunsolo A, Ogunbode C, Middleton J. Anxiety, Worry, and Grief in a Time of Environmental and Climate Crisis: A Narrative Review. *Annual Reviews*. 2021;46:35-58. doi:[10.1146/annurev-environ-012220-022716](https://doi.org/10.1146/annurev-environ-012220-022716).
28. Reyes MES, Carmen BPB, Luminarias MEP, Mangulabnan SANB, Ogunbode CA. An investigation into the relationship between climate change anxiety and mental health among Gen Z Filipinos. *Current Psychology*. 2021;42(9):7448-7456. doi:[10.1007/s12144-021-02099-3](https://doi.org/10.1007/s12144-021-02099-3).
29. Swim JK, Aviste R, Lengieza ML, Fasano CJ. OK Boomer: A decade of generational differences in feelings about climate change. *Global Environmental Change*. 2022;76:102479. doi:[10.1016/j.gloenvcha.2022.102479](https://doi.org/10.1016/j.gloenvcha.2022.102479).
30. Marks E, Atkins E, Garrett JK, et al. Stories of hope created together: A pilot, school-based workshop for sharing eco-emotions and creating an actively hopeful vision of the future. *Frontiers in Psychology*. 2023;13:1076322. doi:[10.3389/fpsyg.2022.1076322](https://doi.org/10.3389/fpsyg.2022.1076322).
31. Aruta JJBR. Environmental psychology in the Philippines: Growth, challenges and prospects. *International Journal of Psychology*. 2023;58(3):217-226. doi:[10.1002/ijop.12896](https://doi.org/10.1002/ijop.12896).
32. Clayton S. Climate anxiety: Psychological responses to climate change. *Journal of Anxiety Disorders*. 2020;74:102263. doi:[10.1016/j.janxdis.2020.102263](https://doi.org/10.1016/j.janxdis.2020.102263).
33. Comtesse H, Ertl V, Hengst SMC, Rosner R, Smid GE. Ecological grief as a response to environmental change: A mental health risk or functional response? *International Journal of Environmental Research and Public Health*. 2021;18(2):734. doi:[10.3390/ijerph18020734](https://doi.org/10.3390/ijerph18020734).
34. Jayathilake H, Annur N. Exploring HRM Practices to Retain Generation-Z in Information Technology Sector: A Systematic Literature Review. *International Journal of Business and Technopreneurship*. 2020;10(1):79-100.
35. Berry HL, Bowen K, Kjellstrom T. Climate change and mental health: A causal pathways framework. *International Journal of Public Health*. 2010;55(2):123-132. doi:[10.1007/s00038-009-0112-0](https://doi.org/10.1007/s00038-009-0112-0).
36. Albrecht G. *Earth Emotions: New Terminology for the Anthropocene*. Cornell University Press; 2019.
37. Crandon TJ, Scott JG, Charlson FJ, Thomas HJ. A social-ecological perspective on climate anxiety in children and adolescents. *Nature Climate Change*. 2022;12(2):123-131. doi:[10.1038/s41558-021-01251-y](https://doi.org/10.1038/s41558-021-01251-y).
38. Pihkala P. The process of eco-anxiety and ecological grief: A narrative review and a new proposal. *Sustainability*. 2022;14(24):16628. doi:[10.3390/su142416628](https://doi.org/10.3390/su142416628).
39. Johnson B. Toward a New Classification of Nonexperimental Quantitative Research. *Educational Researcher*. 2001;30(2):3-13. doi:[10.3102/0013189X030002003](https://doi.org/10.3102/0013189X030002003).

40. Ajwad MI, Kurukulasuriya P, Basist A, et al. Cross-Sectional Analyses of Climate Change Impacts | Policy Research Working Papers. *Policy Research Working Papers*. 2022;350. doi:[10.1596/1813-9450-3350](https://doi.org/10.1596/1813-9450-3350).
41. Belli G. Nonexperimental quantitative research. In: Lapan SD, Quartaroli MT, eds. *Research essentials: An introduction to designs and practices*. San Francisco, CA: Jossey-Bass; 2009:59-77.
42. Bashan D, Colléony A, Shwartz A. Urban versus rural? The effects of residential status on species identification skills and connection to nature. *People and Nature*. 2021;3(2):347-358. doi:[10.1002/pan3.10176](https://doi.org/10.1002/pan3.10176).
43. Boluda-Verdu I, Senent-Valero M, Casas-Escolano M, Matijasevich A, Pastor-Valero M. Fear for the future: Eco-anxiety and health implications, a systematic review. *Journal of Environmental Psychology*. 2022;84:1019024. doi:[10.1016/j.jenvp.2022.101904](https://doi.org/10.1016/j.jenvp.2022.101904).
44. Clayton SD, Pihkala P, Wray B, Marks E. Psychological and emotional responses to climate change among young people worldwide: Differences associated with gender, age, and country. *Sustainability*. 2023;15(4):3540. doi:[10.3390/su15043540](https://doi.org/10.3390/su15043540).
45. Duron-Ramos MF, Collado S, García-Vázquez FI, Bello-Echeverría M. The role of urban/rural environments on Mexican children's connection to nature and pro-environmental behavior. *Frontiers in Psychology*. 2020;11:514. doi:[10.3389/fpsyg.2020.00514](https://doi.org/10.3389/fpsyg.2020.00514).
46. Gifford R, Nilsson A. Personal and social factors that influence pro-environmental concern and behaviour: A review. *International Journal of Psychology*. 2014;49(3):141-157. doi:[10.1002/ijop.12034](https://doi.org/10.1002/ijop.12034).
47. Kankawale S, Niedzwiedz CL. Eco-anxiety among Children and Young People: Systematic Review of Social, Political, and Geographical determinants. *medRxiv (Cold Spring Harbor Laboratory)*. 2023; doi:[10.1101/2023.12.19.23300198](https://doi.org/10.1101/2023.12.19.23300198).
48. Xiao C, McCright AM. Gender differences in environmental concern: Revisiting the institutional trust hypothesis in the USA. *Environment and Behavior*. 2015;47(1):17-37. doi:[10.1177/0013916513491571](https://doi.org/10.1177/0013916513491571).
49. Albrecht G. Chronic Environmental Change: Emerging 'Psychoterratic' Syndromes. In: Weissbecker I, ed. *Climate Change and Human Well-Being*. New York, NY: Springer; 2011:43-56.
50. McGee R. Welch's t-test: The robust default for comparing two groups in psychological research. *Sociology & Social Policy*. 2025; 2(2):52-58. doi:[10.26855/ssp.2025.09.003](https://doi.org/10.26855/ssp.2025.09.003).
51. Sandelowski M. What's in a name? Qualitative description revisited. *Research in Nursing & Health*. 2010;33(1):77-84. doi:[10.1002/nur.20362](https://doi.org/10.1002/nur.20362).
52. Mashuri S, Sarib M, Alhabsyi F, Ruslin R. *Semi-structured Interview: A Methodological Reflection on the Development of a Qualitative Research*. ResearchGate; 2022. doi:[10.9790/7388-1201052229](https://doi.org/10.9790/7388-1201052229).
53. Creswell J. *Qualitative inquiry and research design: Choosing among five traditions*. Sage; 1998.
54. Morse J. Designing funded qualitative research. In: Denzin N, Lincoln Y, eds. *Handbook for qualitative research*. Thousand Oaks, CA: Sage; 1994:220-235.
55. Guest G, Bunce A, Johnson L. How many interviews are enough? An experiment with data saturation and variability. *Field Methods*. 2006;18(1):59-82. doi:[10.1177/1525822X05279903](https://doi.org/10.1177/1525822X05279903).
56. Tan V. Sleeping with My Abuser: A Qualitative Study on the Development of Transferred Aggression Expression among Battered Women. *Journal of Psychology & the Behavioral Sciences*. 2018;4(1): doi:[10.22492/ijpbs.4.1.06](https://doi.org/10.22492/ijpbs.4.1.06).
57. Serrano JO, Reyes MES. Bending not breaking: coping among Filipino University students experiencing psychological distress during the Global Health Crisis. *Current Psychology*. 2023;42:28857-28867. doi:[10.1007/s12144-022-03823-3](https://doi.org/10.1007/s12144-022-03823-3).
58. Villamor NJ, de Guzman AB, Matienzo ET. The Ebb and Flow of Filipino First-Time Fatherhood Transition Space: A Grounded Theory Study. *American Journal of Men's Health*. 2016;10(6):51-62. doi:[10.1177/1557988315604019](https://doi.org/10.1177/1557988315604019).
59. Ahmed SK. Sample size for saturation in qualitative research: Debates, definitions, and strategies. *Journal of Medicine, Surgery, and Public Health*. 2025;5:100171. doi:[10.1016/j.glmedi.2024.100171](https://doi.org/10.1016/j.glmedi.2024.100171).
60. Kiger ME, Varpio L. Thematic analysis of qualitative data: AMEE Guide No. 131. *Medical Teacher*. 2020;42(8):1-9. doi:[10.1080/0142159X.2020.1755030](https://doi.org/10.1080/0142159X.2020.1755030).
61. Braun V, Clarke V. Using thematic analysis in psychology. *Qualitative Research in Psychology*. 2006;3(2):77-101. doi:[10.1191/1478088706qp063oa](https://doi.org/10.1191/1478088706qp063oa).
62. Carcary M. The Research Audit Trail: Methodological Guidance for Application in Practice. *The Electronic Journal of Business Research Methods*. 2020;18(2):166-177. doi:[10.34190/JBRM.18.2.008](https://doi.org/10.34190/JBRM.18.2.008).
63. McKim C. Meaningful Member-Checking: A Structured Approach to Member-Checking. *American Journal of Qualitative Research*. 2023;7(2):41-52.
64. Palaganas EC, Sanchez MC, Molintas MP, Caricativo RD. Reflexivity in Qualitative Research: A Journey of Learning. *The Qualitative Report*. 2017;22(2):426-438. doi:[10.46743/2160-3715/2017.2552](https://doi.org/10.46743/2160-3715/2017.2552).
65. Jamieson MK, Govaart GH, Pownall M. Reflexivity in quantitative research: A rationale and beginner's guide. *Social and Personality Psychology Compass*. 2023;17(4):e12735. doi:[10.1111/spc3.12735](https://doi.org/10.1111/spc3.12735).
66. Wilkinson S. The role of reflexivity in feminist psychology. *Women's Studies International Forum*. 1988;11(5):493-502. doi:[10.1016/0277-5395\(88\)90024-6](https://doi.org/10.1016/0277-5395(88)90024-6).
67. Willig C. *Introducing qualitative research in psychology*. McGraw-Hill Education; 2013.

68. Galway LP, Field E. Climate emotions and anxiety among young people in Canada: A national survey and call to action. *The Journal of Climate Change and Health*. 2023;9:100204. doi:[10.1016/j.joclim.2023.100204](https://doi.org/10.1016/j.joclim.2023.100204).
69. Albrecht G, Sartore GM, Connor L, et al. Solastalgia: The distress caused by environmental change. *Australasian Psychiatry*. 2007;15(sup1):S95-S98. doi:[10.1080/10398560701701288](https://doi.org/10.1080/10398560701701288).
70. Ellis NR, Albrecht GA. Climate change threats to family farmers' sense of place and mental wellbeing: A case study from the Western Australian Wheatbelt. *Social Science & Medicine*. 2017;175:161-168. doi:[10.1016/j.socscimed.2017.01.009](https://doi.org/10.1016/j.socscimed.2017.01.009).
71. Vicente-Molina MA, Fernández-Sainz A, Izagirre-Olaizola J. Does gender make a difference in pro-environmental behavior? The case of the Basque Country University students. *Journal of Cleaner Production*. 2018;176:89-98. doi:[10.1016/j.jclepro.2017.12.079](https://doi.org/10.1016/j.jclepro.2017.12.079).
72. Holmes M, Natalier K, Leahy CP. Unsettling maternal futures in climate crisis: towards cohabitability? *Families, Relationships and Societies*. 2023;12(3):357-373. doi:[10.1332/204674321X16621119776374](https://doi.org/10.1332/204674321X16621119776374).
73. Peters S, Striessnig E, Trimarchi A, Testa MR, Nitsche N. Too worried about the environment to have children? Or more worried about the environment after having children? *Population and Environment*. 2025;47:31. doi:[10.1007/s11111-025-00501](https://doi.org/10.1007/s11111-025-00501).
74. Eneji CVOE, Onnoghen NU, Acha JO, Diwa JB. Climate change awareness, environmental education and gender role burdens among rural farmers of Northern Cross River State, Nigeria | Emerald Insight. *International Journal of Climate Change Strategies and Management*. 2020;13(4/5):397-415. doi:[10.1108/IJCCSM](https://doi.org/10.1108/IJCCSM).
75. Velasco P, Richardson L. Ecological Grief as a Shared Emotion. *Emotion Review*. 2025;18. doi:[10.1177/17540739251379796](https://doi.org/10.1177/17540739251379796).
76. Sapovadia VK. *The Emotional Ecology of Climate Action: Harnessing Collective Grief and Hope*; 2025. Available at SSRN 5291955.
77. Cunsolo A, Landman K. *Mourning nature: Hope at the heart of ecological loss and grief*. McGill-Queen's Press-MQUP; 2017.
78. Colléony A, Levontin L, Shwartz A. Promoting meaningful and positive nature interactions for visitors to green spaces. *Conservation Biology*. 2020;34(6):1373-1382. doi:[10.1111/cobi.13624](https://doi.org/10.1111/cobi.13624).
79. Pohl L, Helbrecht I. The love of nature: Imaginary environments and the production of ontological security in postnatural times. *Geo: Geography and Environment*. 2022;9(1):e00106. doi:[10.1002/geo2.106](https://doi.org/10.1002/geo2.106).
80. Mann S, McKay T, Gonzales G. Climate change-related disasters & the health of LGBTQ+ populations. *The Journal of Climate Change and Health*. 2024;18:100304. doi:[10.1016/j.joclim.2024.100304](https://doi.org/10.1016/j.joclim.2024.100304).
81. Nielsen KS, Nicholas KA, Creutzig F, Dietz T, Stern PC. The role of high-socioeconomic-status people in locking in or rapidly reducing energy-driven greenhouse gas emissions. *Nature Energy*. 2021;6(11):1011-1016. doi:[10.1038/s41560-021-00900-y](https://doi.org/10.1038/s41560-021-00900-y).